PassFrame tuition

Mixed Mental Maths

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 91

2. C — 9/20 = 45/100 = 45%

3. 85,000

4. 6,812

5. B — 54 = 9 × 6

6. 40 (72÷9=8, 8×5=40)

7. 270 minutes

8. £6.35

9. B and D (6 and 8 both divide exactly into 48; 5, 7 and 9 do not)

10. 532

11. 49 (84÷12=7, 7×7=49)

12. D — A reflex angle is greater than 180°

13. 136

14. 8°C (−14+22=8)

15. 2,625

Standard

1. 144

2. 87.5%

3. A — 65/100 = 13/20

4. 160,000

5. 14

6. 3,750 ml

7. C — 24 is divisible by both 6 and 8

8. 36 (63÷7=9, 9×4=36)

9. 56 cm (2×(18.5+9.5)=56)

10. 11:25

11. A, B, D and E (10/16, 15/24, 62.5% and 0.625 are all equivalent to 5/8; 5/16 is not)

12. 2,880

13. 15 (14+22+9+17+13=75; 75÷5=15)

14. B — 0.625 > 0.62 > 0.6

15. 90 km/h (396÷4.4=90)

Standard / Hard

1. 72 (108÷9=12, 12×7=84, 84−12=72)

2. C — 18% of 250 = 45; 250 − 45 = 205

3. 322 (389 − 67 = 322)

4. CDLXXVI (CD + LXX + VI)

5. 60 cm (7.2 m ÷ 12 = 0.6 m = 60 cm)

6. C — 195÷13=15

7. 650 g (450÷9=50 g per person; 50×13=650 g)

8. -8°C (−19+26=7; 7−15=−8)

9. A, B, D and E (7/8=87.5%, 11/20=0.55, 0.32=32% and 17/25=0.68 are all correct; 5/9 (≈0.556) is actually less than 2/3 (≈0.667), not greater)

10. 264 cm² (24×11=264)

11. 11:25

12. C — Each term is multiplied by 3: 162×3=486

13. £12.55 (£40.00 − £27.45 = £12.55)

14. 63 (98÷14=7, 7×9=63)

15. 5 hours (45÷15=3 hours, plus 40÷20=2 hours, total 5 hours)

Homework — Answers

Easy / Standard

1. 98

6. 36 (81÷9=9, 9×4=36)

11. 40 (88÷11=8, 8×5=40)

2. C — 3/8 = 0.375 = 37.5%

3. 62,000

4. 8,441

5. A — 77 = 11 × 7

7. 225 minutes

8. £7.15

9. C, D and E (10, 12 and 15 all divide exactly into 60; 7 and 8 do not)

10. 654

12. C — An obtuse angle is between 90° and 180°

13. 114

14. 8°C (−21+29=8)

15. 2,515

Standard

1. 130

2. 32.5%

3. A — 85/100 = 17/20

4. 240,000

5. 17

6. 650 g

7. B — 63 is divisible by both 7 and 9

8. 75 (90÷6=15, 15×5=75)

9. 72 cm (2×(22.5+13.5)=72)

10. 16:05

11. A, B, D and E (6/16, 9/24, 37.5% and 0.375 are all equivalent to 3/8; 3/16 is not)

12. 4,340

13. 20 (18+27+12+21+22=100; 100÷5=20)

14. B — 0.785 > 0.78 > 0.7

15. 87 km/h (522÷6=87)

Standard / Advanced

1. 75 (144÷12=12, 12×5=60, 60+15=75)

2. B — 22% of 320 = 70.40; 320 − 70.40 = 249.60

3. 368 (461 − 93 = 368)

4. DCXLIX (DC + XL + IX)

5. 60 cm (8.4 m ÷ 14 = 0.6 m = 60 cm)

6. C — 255÷17=15

7. 850 g (600÷12=50 g per person; 50×17=850 g)

8. -11°C (−27+35=8; 8−19=−11)

9. A, B, D and E (9/10=90%, 7/25=0.28, 0.44=44% and 23/50=0.46 are all correct; 7/12 (≈0.583) is actually less than 5/8 (0.625), not greater)

10. 448 cm² (32×14=448)

11. 12:25

12. C — Each term is multiplied by 3: 216×3=648

13. £21.25 (£60.00 − £38.75 = £21.25)

14. 88 (120÷15=8, 8×11=88)

15. 5 hours (60÷20=3 hours, plus 56÷28=2 hours, total 5 hours)